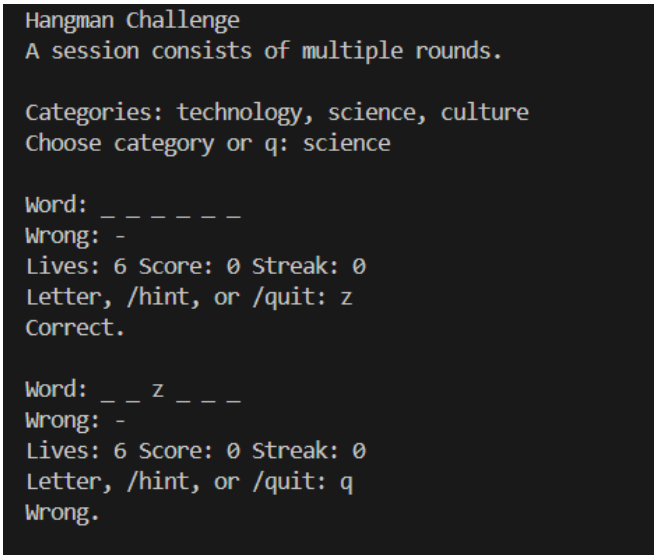
SE LAB LAB-4
NAME:VARUN SAHU
SRN:PES2UG24CS576
SECTION:J

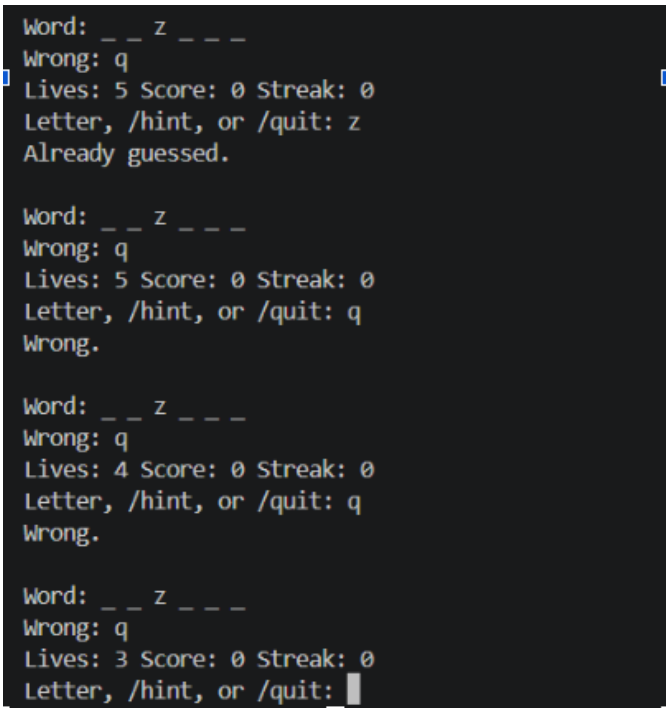
Objective
Use Vibe Coding tools to fix the broken code and add features to the same. Learn how to prompt to get the job done in under three to four attempts. Clone a repo assigned to you, make changes and push to your repo in your personal account.

Repo: https://github.com/varunsahu5002/SE_ARCHITECTURE_LAB4.git

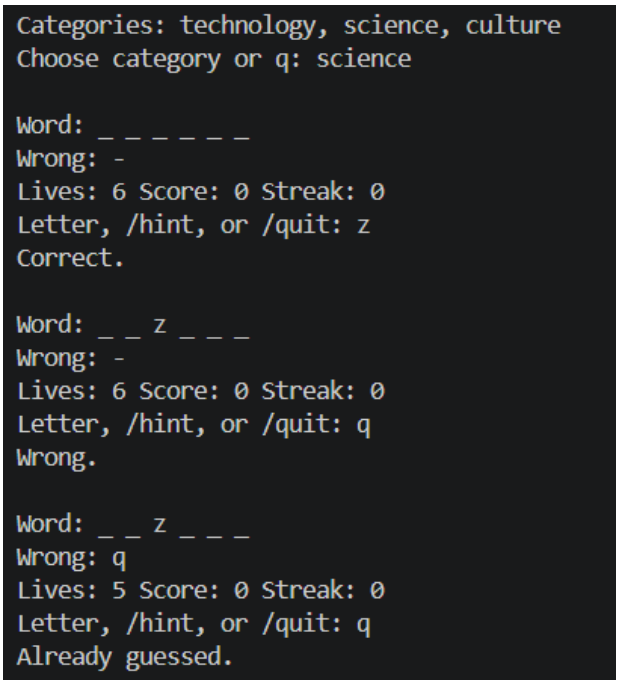
Task 1 — Guess-state correctness

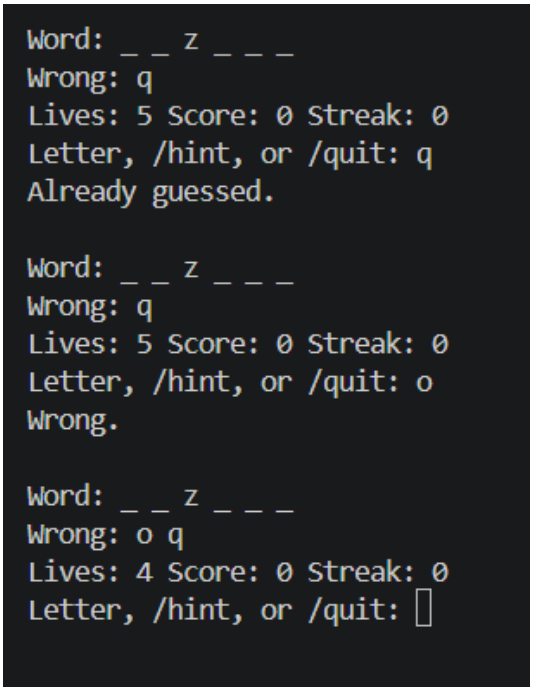
Correct repeated-guess handling so a previously attempted wrong letter cannot consume another life, and a previously accepted correct letter cannot be counted as a new guess.
Done when: each distinct letter affects the round exactly once.
Before:





After:



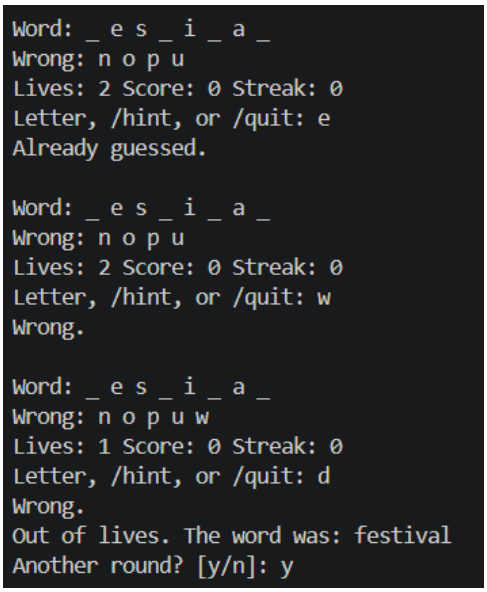


Task 2 — Complete the session model

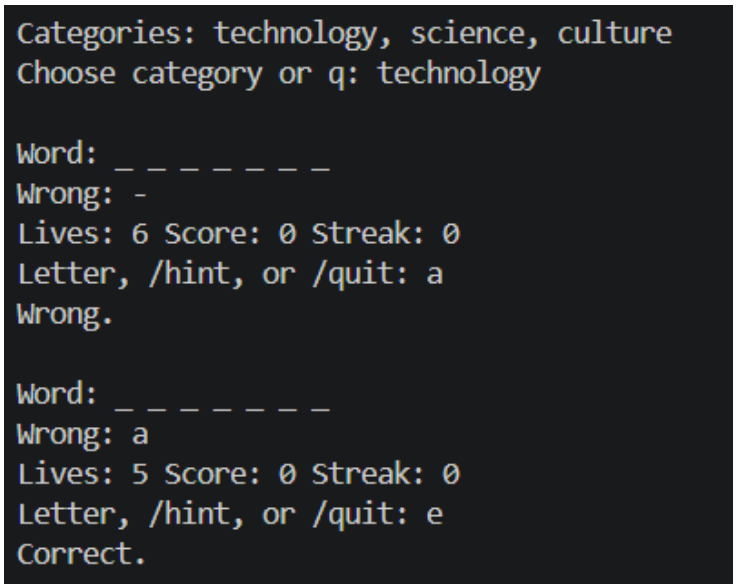
Integrate the supplied session-statistics component so rounds played, rounds won, and best streak are tracked correctly across multiple rounds.
Done when: starting a new round resets only round-specific state; session statistics continue across the whole run.

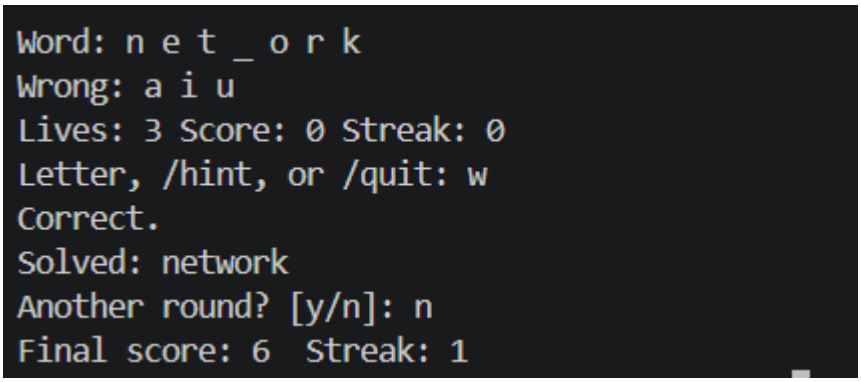
Before:

Round 1



round 2:

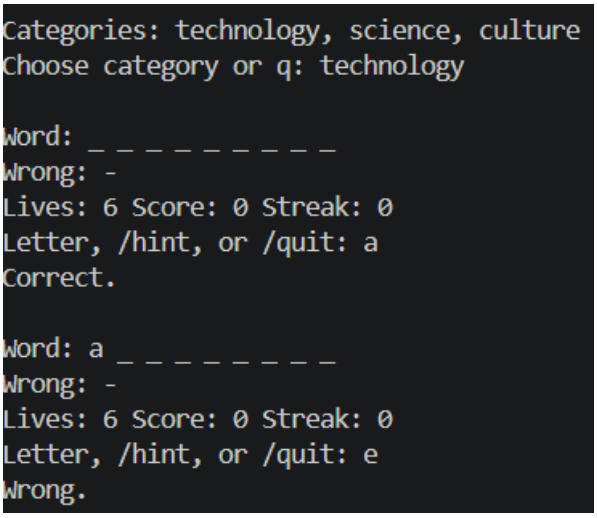




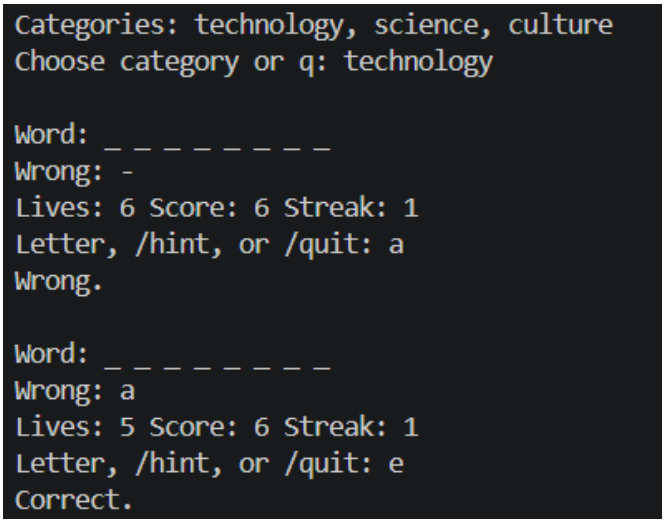
Bug here is not displaying match stats.

After:

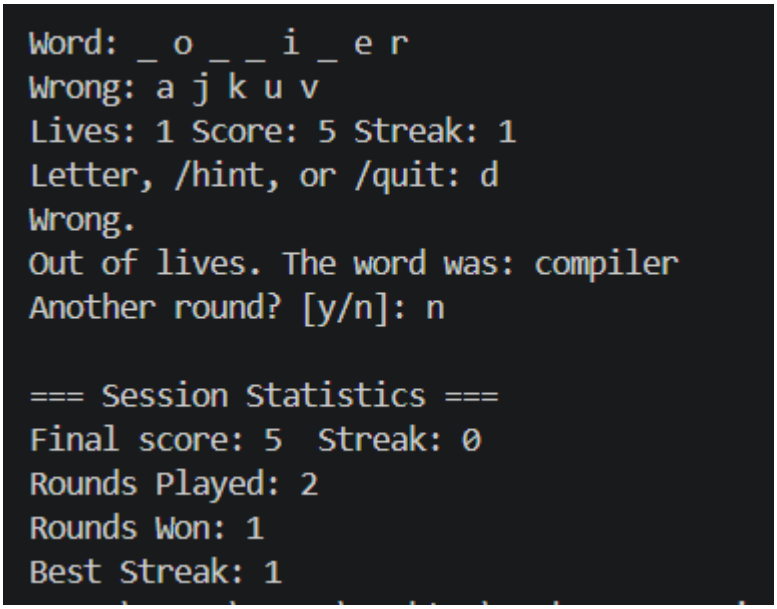
Round1:



round 2:



Bug fix:



The match stats have been displayed by making changes to [game.py]

Task 3 — Difficulty and scoring

Add difficulty choices that alter the available lives and scoring. Preserve category selection and make hint usage affect scoring consistently.
Done when: difficulty changes round rules without leaking state between rounds.

Before:
The game hardcodes your starting lives to 6.
It never asks you to choose a difficulty.
It always subtracts exactly 1 point when you use a hint, and the points you earn for winning never scale.

```
Categories: technology, science, culture
Choose category or q: culture

Word: _ _ _ _ _ _ _ _ _
Wrong: -
Lives: 6 Score: 13 Streak: 2
Letter, /hint, or /quit: a
Correct.

Word: _ a _ _ _ a _ _
Wrong: -
Lives: 6 Score: 13 Streak: 2
Letter, /hint, or /quit: e
Correct.
```

```
Word: _ a _ _ u a _ e
Wrong: i o
Lives: 4 Score: 13 Streak: 2
Letter, /hint, or /quit: /hint
A system used for communication.

Word: _ a _ _ u a _ e
Wrong: i o
Lives: 4 Score: 12 Streak: 2
Letter, /hint, or /quit: l
Correct.
```

After:
Difficulty can be chosen
When easy is chosen lives are 8(no hardcode
Of 6 lives)
when difficulty: difficult is selected upon using /hint
Score drops by 3 instead of 1
Lives start with 4 instead of hardcoded 6 lives

```
Categories: technology, science, culture
Choose category or q: technology
Choose difficulty (easy/medium/hard): easy

Word: _ _ _ _ _ _ _ _
Wrong: -
Lives: 8 Score: 0 Streak: 0
Letter, /hint, or /quit: a
Wrong.
```

```
Word: _ e _ _ _ a _ e
Wrong: -
Lives: 4 Score: 6 Streak: 1
Letter, /hint, or /quit: d
Wrong.

Word: _ e _ _ _ a _ e
Wrong: d
Lives: 3 Score: 6 Streak: 1
Letter, /hint, or /quit: a
Already guessed.

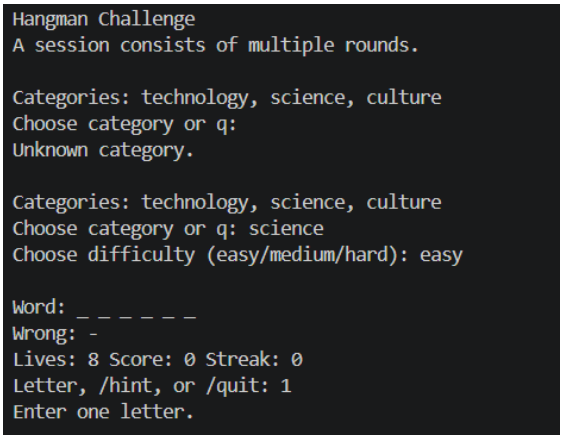
Word: _ e _ _ _ a _ e
Wrong: d
Lives: 3 Score: 6 Streak: 1
Letter, /hint, or /quit: /hint
Something inherited from previous generations.

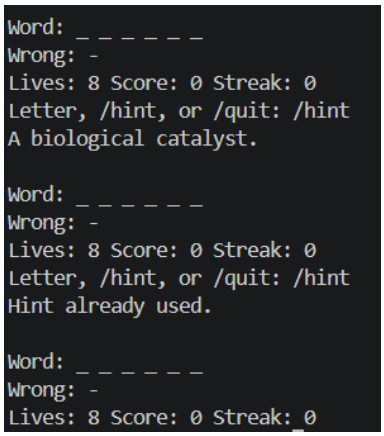
Word: _ e _ _ _ a _ e
Wrong: d
Lives: 3 Score: 3 Streak: 1
Letter, /hint, or /quit: h
Correct.
```

Task 4 — Robust input and feedback

Improve command handling for invalid letters, repeated commands, category selection, and hint usage. Feedback should describe actual player actions once.
Done when: malformed input never changes game state and feedback is not duplicated.

Before:
Invalid Category Handling: Typing an incorrect category forces the entire main menu to reprint instead of simply asking for the input again.
Buried Feedback: Error messages (like "Enter one letter") are pushed out of sight because the game immediately reprints the full board right below them.
Unprotected Inputs: Hitting enter on an empty prompt or typing symbols causes the game to cycle pointlessly, cluttering the terminal history.





After:
Clean Menus: Invalid category or difficulty choices now smoothly ask for the input again without reprinting the entire menu.
Targeted Feedback: Error messages are held and displayed clearly right above the next input prompt (using `>>`) instead of getting buried by the board.
Strict Validation: The game properly catches empty inputs, numbers, and symbols, preventing the board from pointlessly cycling.

